

CRUX

failure discovery → repair → qualification

for contact-rich robot manipulation — end to end on one AMD Radeon GPU

0/32 → 12/32 reached seating · +37.5 pp · exact McNemar $p = 0.0005$

replicated identically on a second independent suite · release gate: APPROVED

first completed episodes on camera — success 1/32, honestly labelled not yet significant

THE DISCOVERY CAMPAIGN

17 matched sweep rounds

~1,000 batched episodes

8 mechanisms isolated

lift-then-translate transport

grip the connector link

-56 N clamp → sub-mm alignment

re-observed grasp retries

...and 5 falsified repair families

THE METRIC BUG

Five independent seating methods

all stalled at 12.0-13.4 mm.

That floor WAS the fully-seated position: the metric measured the connector's trailing joint - success was impossible by design.

Found by falsification. Fixed.

SCALE, ON ONE RADEON

~200k-293k env-steps / second

4,096 batched environments

GPU 100% busy under live load

96-episode qualification: 114 s

Radeon PRO W7900 - gfx1100

ROCm 7.2.1 - Genesis 1.3.1

torch 2.13.0+rocm7.2

HONESTY IS THE PRODUCT

every number recomputed from

hash-verified raw episodes

a non-replicating claim was

withdrawn in writing

failed episodes never deleted

2 retractions, on the record

3 Genesis issues filed upstream

212 CPU tests - no GPU needed

VERIFY IT YOURSELF — CPU ONLY

```
git clone https://github.com/Enoch208/Crux && cd Crux && uv sync
```

```
uv run pytest -q # 212 tests, ~1 s
```

```
uv run crux validate evidence/manifest.json # 9/9 checks passed
```

Tamper with one byte of an episode file and it fails. That is the point.

[enoch208.github.io/Crux](https://github.com/Enoch208/Crux)

github.com/Enoch208/Crux · [docs/technical-report.md](#) · 3.5-min narrated demo in-repo